

Predicting IMDb Film Ratings

Math 748 — Final Project

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Outline

- 1 Introduction
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- 3 Methods & Results
- 4 Comparison & Conclusions

- Film success is quantified by **quality** and **popularity**
- IMDb average rating captures one population's sentiment toward a film
- **Goal:** Determine which film features drive average rating, and compare models

Models Used

- Linear Regression
- Ridge & Lasso
- Support Vector Regression
- Random Forests

Source: IMDb (downloaded Feb 3, 2026)

- `title.basics` — metadata (title, genre, runtime)
- `title.ratings` — average rating, number of votes, (other elements that were removed)
- Merged on `tconst`; restricted to **feature films only**

Dataset Size

- `title.basics`: 12,263,801 rows
- `title.ratings`: 1,631,482 rows
- Final model dataset: feature films, 1970–2025

Data Preprocessing

- ➊ **Remove irrelevant columns:** `titleType` $\neq$ movie, `isAdult`, `originalTitle`, `endYear`
- ➋ **Filter years:** Remove films before 1970 (high missingness, lost films, classic Hollywood)
- ➌ **Handle missingness:** Drop rows missing `runtimeMinutes` or `genres`
- ➍ **One-hot encode genres:** Many films span multiple genres; remove genres $< 1\%$
- ➎ **Runtime filtering:** Keep films between 60–300 minutes (Under an hour usually not considered feature length)
- ➏ **Log transform votes:** $\log\text{Votes} = \log(\text{numVotes})$ to reduce skew
- ➐ **Baseline genre:** Remove Drama as reference category in linear models

Missing Values

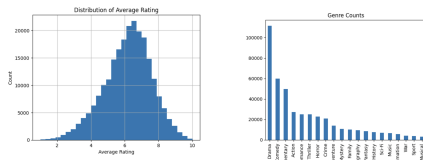
Variable	Missing	% Missing
tconst	0	0.00
primaryTitle	2	0.00
startYear	35	0.01
runtimeMinutes	32,817	9.83
genres	11,051	3.31
averageRating	0	0.00
numVotes	0	0.00

Missingness in runtimeMinutes and genres is much higher before 1970. Films from early and classic Hollywood are often lost or less updated in databases.

Response Variable & Key Features

Response: averageRating

- Approximately normal, slightly left-skewed
- Peaks around **6.3**
- Rarely drops very low — users do not often rate films lower than 4/5



Genre Distribution (top)

Drama ≫ Comedy > Documentary > Thriller
> ...

Predictors:

- startYear
- runtimeMinutes
- Genre indicators (one-hot)
- logVotes (explored separately)

Removed genres < 1%: Western, News, Reality-TV, Talk-Show, Game-Show (likely misclassified TV content).

Linear Regression (Baseline)

- 80/20 train/test split, seed = 748
- Minimizes OLS squared error
- **RMSE = 1.233**, $R^2 = 0.227$
- Within ~ 1 rating point on a 1–10 scale ($\sim 10\%$ off)
- Low R^2 likely due to noise and the massive variety of films

Key coefficients:

- *Positive*: Documentary, Animation, Music
- *Negative*: Horror, Sci-Fi, Action
- runtimeMinutes positive but small

Including logVotes

Adding logVotes produced **identical** RMSE = 1.233 and $R^2 = 0.227$. Its coefficient was -0.011 — negligible effect. Excluded from final models.

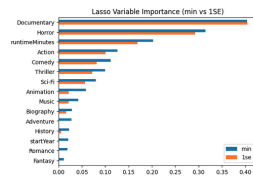
Ridge & Lasso Regression

Ridge (L2):

- 5-fold CV on log-spaced λ grid
- **RMSE = 1.234** — marginal increase over OLS

Lasso (L1):

- $\lambda_{\min} = 0.001$: RMSE = 1.234, **19 features**
- $\lambda_{1se} = 0.029$: RMSE = 1.238, **11 features**



Most Important Predictors (Lasso)

Documentary and Horror dominate both λ settings, with runtimeMinutes close behind — consistent with linear regression findings.

The λ_{1se} model trades minimal accuracy for a much simpler, more interpretable model. Given comparable RMSE, simpler is preferred.

Support Vector Regression (SVR)

Three models tested; Drama included (interpretation less critical):

Model	Kernel	RMSE
Linear SVR	Linear	1.229
RBF SVR (untuned)	RBF	1.222
RBF SVR (tuned)	RBF	1.217

- RBF models trained on 10,000-row subsample (computational cost)
- Tuning via `GridSearchCV` with 3-fold CV over C , γ , ϵ
- All SVR models outperform OLS, but gains are small
- **Tradeoff:** Complexity sacrifices interpretability

Random Forests

Untuned RF (`n_estimators=200`):

- **RMSE = 1.304** — worst of all models
- Top features: `runtimeMinutes`, `startYear` (genres missing)

Tuned RF (`max_depth=15`, `min_samples_leaf=5`):

- **RMSE = 1.200** — best of all models
- Limiting depth reduced overfitting
- Top features: `Documentary`, `Horror`, `runtimeMinutes`

Tuned RF Feature Importance (top 5)

Feature	Importance
Documentary	0.385
Horror	0.179
runtimeMinutes	0.158
startYear	0.092
Drama	0.069

Model Performance Summary

Model	Test RMSE
Linear Regression	1.233
Ridge (CV)	1.234
Lasso ($\lambda_{\min}$)	1.234
Lasso (λ_{1se})	1.238
Linear SVR	1.229
RBF SVR (untuned)	1.222
RBF SVR (tuned)	1.217
Random Forest (untuned)	1.304
Random Forest (tuned)	1.200

*RBF SVR models trained on 10,000-row subsample.

Key Findings

Model Performance:

- Tuned Random Forest achieves best RMSE (1.200)
- Most models cluster tightly around 1.22–1.24
- Ridge/Lasso offer no accuracy gain over OLS, but improve interpretability and variable selection
- Untuned RF is the worst — depth control needed

Variable Importance:

- **Documentary** is the strongest positive predictor across all models
- **Horror** is the strongest negative predictor
- `runtimeMinutes` consistently important — longer films rated higher. Makes sense.
- `startYear` matters in tree models; IMDb users show era bias
- `logVotes` adds no predictive power

Conclusions & Future Work

Conclusions:

- Genre and runtime are the primary drivers of `averageRating`
- The tuned Random Forest is the best predictive model ($\text{RMSE} = 1.200$)
- For interpretation, Lasso with $\lambda_{1\text{se}}$ gives a clean, parsimonious model with comparable accuracy
- Underlying signal is stable — top predictors agree across all models

Future Work

- Explore `logVotes` more deeply — potential nonlinear interaction with genre
- Runtime binning (short/medium/long)
- Expand to director/actor features

Thank You

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